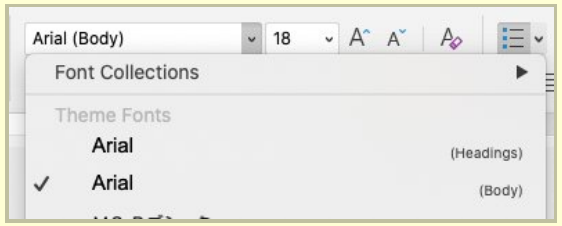
# >>>>> Before you begin <<<<<

## Capitalization guidelines

- Use **title case** for presentation titles
  - All words are capitalized except for prepositions, conjunctions, and articles
- Use **sentence case** for slide titles, section titles, and bullets
  - Just the first word in a text string is capitalized
- Always **capitalize proper nouns**, including Esri product names as noted in the Product Naming Guide

## Font guidelines

- Always pick **Arial** [Headings] or [Body] from the **top** of the font list under Theme Fonts



# >>>>>  Before you begin  <<<<<



## Less is more

- Avoid paragraphs and long sentences
- Use the 5x5 rule — max of 5 lines, 5 words each
- One idea per slide



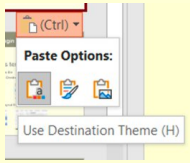
## Presentation structure

- Content in longer presentations should be grouped using the following structure:
  - Sections > Topics > Subtopics
  - Note: A section may include multiple topics

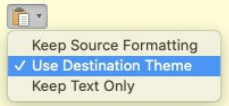
# >>>>> Before you begin <<<<<

## How to apply this template

1. Copy existing slides into this file
2. When pasting, choose Use Destination Theme




*Windows*

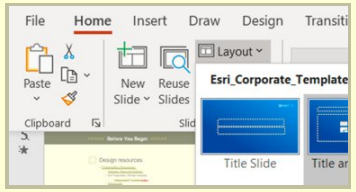



*Mac*

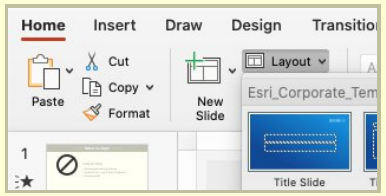



*Web*

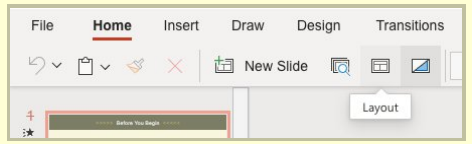
3. For consistent title placement, you may need to select a layout from the Layout menu




*Windows*




*Mac*




*Web*

# >>>>> Before you begin <<<<<

## Design resources

- Icons
  - From the Home ribbon above, click Esri Icons

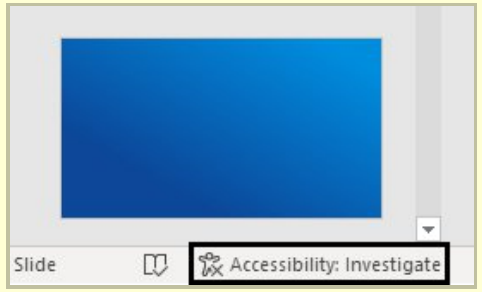



- Presentation Resources [link]
  - Diagrams, Tables and Timelines [link]
  - Esri Design Template [link]
  - Globes [link]
  - Backgrounds [link]
- Strategic Presentations [link]
- Stock photos
  - Adobe Stock images are available. See the details
  - **Do not use** stock images from Microsoft.

# >>>>>  Before you begin  <<<<<

## Accessibility [link]

- Check accessibility while designing slides with the Review tab or in the lower left-hand corner
- Add **alternative text** and mark graphical images as decorative
- Review slide **reading order**
- Create **unique slide titles**






# Performant maps, Open data

## Optimizing large municipal queries with MapLibre

George Owen

# Agenda

- Municipal open data
- Browser-based data rendering
- Optimizing GeoJSON queries

# Municipal open data

# Finding open data

Focus of today's talk

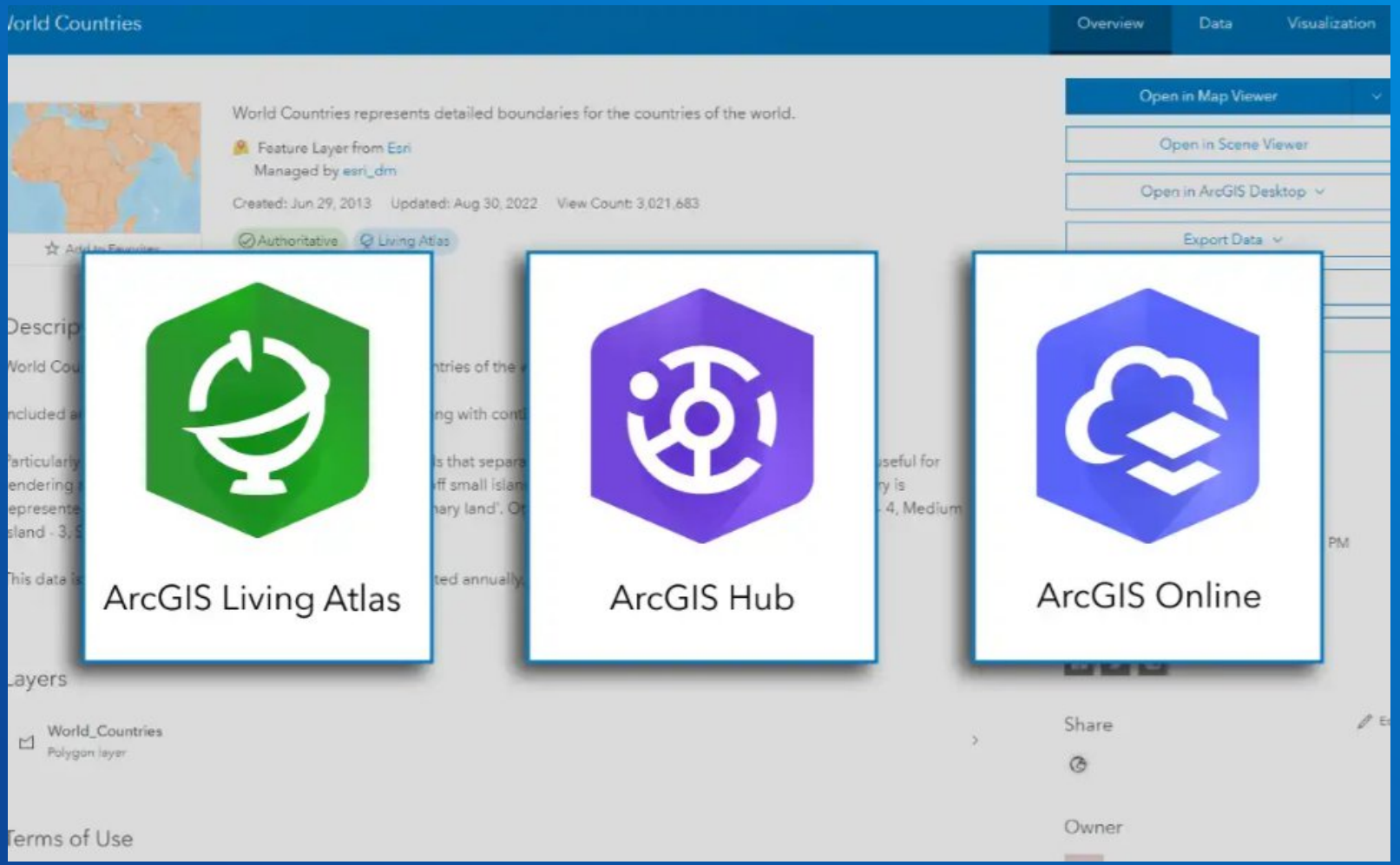
- **ArcGIS Hub:** Municipal data, local gov't
- Today's study area: **City of Sacramento**
- Data sources used in today's talk:
  - City of Sacramento Open Data Hub
  - Sacramento County Open Data Hub
  - ArcGIS Living Atlas, Esri Demographics*



# Other open data sources

- https://livingatlas.arcgis.com/
- https://hub.arcgis.com/

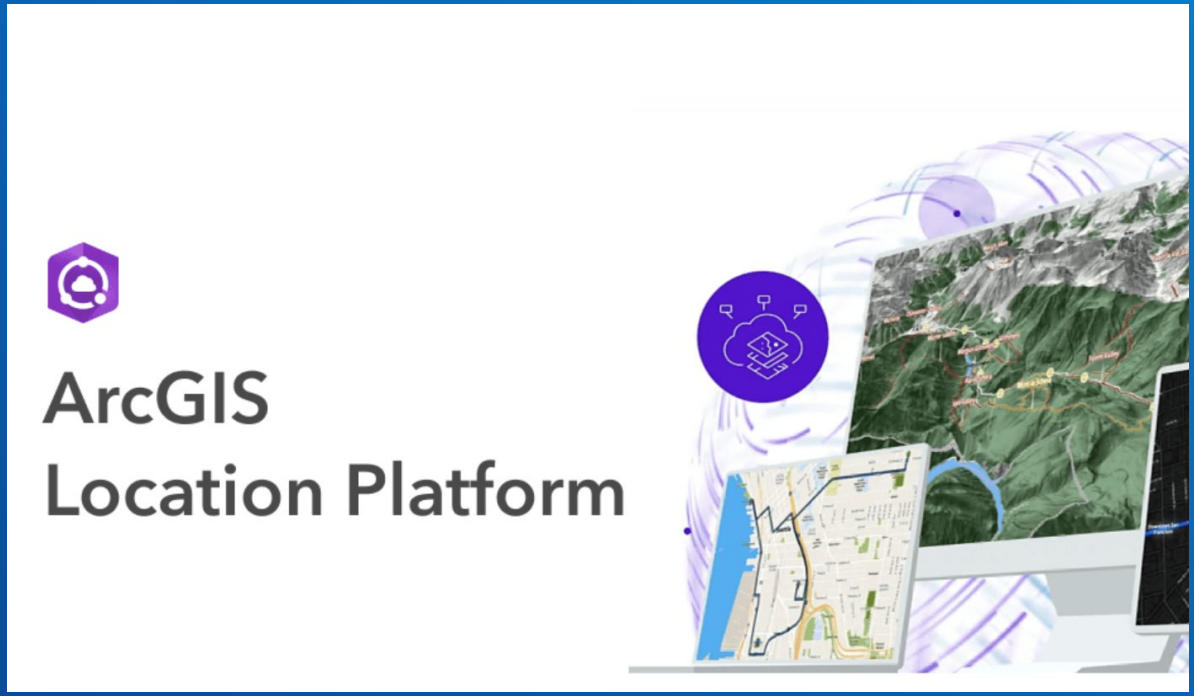
- All free
- Worth checking your area



# Accessing ArcGIS open data

## Use ArcGIS Location Platform

- Free and open source
- Make a free account: ArcGIS Location Platform
- Access Hub data, Living Atlas data
- Also get data hosting, platform services



# Features: Open data formats

- A world of wonderful standards!
- JSON-FG, GeoParquet, Vector tiles, GeoPackage, Topo-JSON, PMTiles, KMZ.....

- Reality: **GeoJSON everywhere**

- Esri feature services return GeoJSON.
  - Among other, worse options



# Browser-based data rendering

# MapLibre GL JS

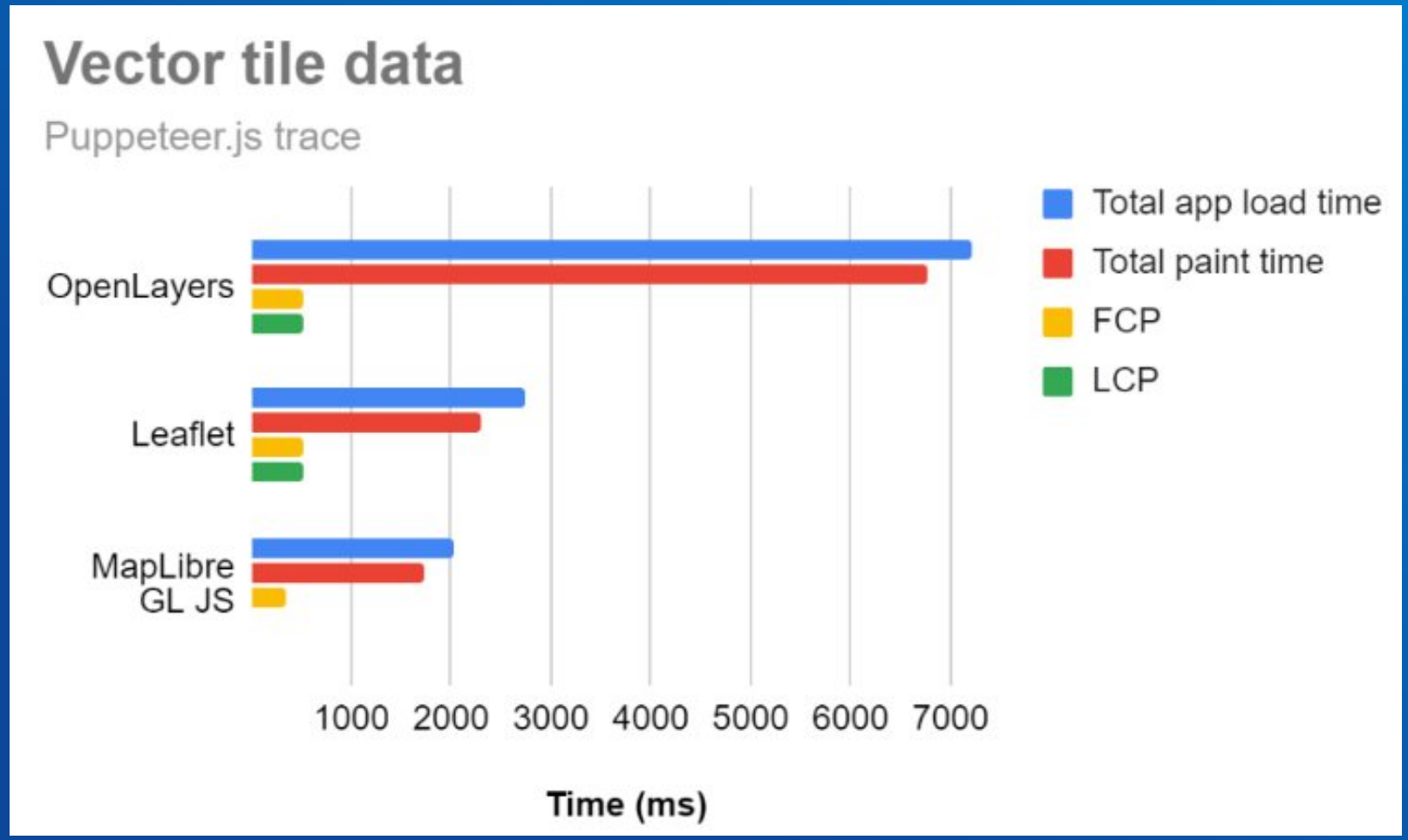
https://maplibre.org/

- Open-source fork of Mapbox v1, 2020
- MapLibre ecosystem: Native SDK, tile server, and more
- Wonderful alternative to Mapbox

- Strong industry support
- Active community
- Growing in popularity



# MapLibre GL JS is performant




- Source: my 2023 FOSS4G talk

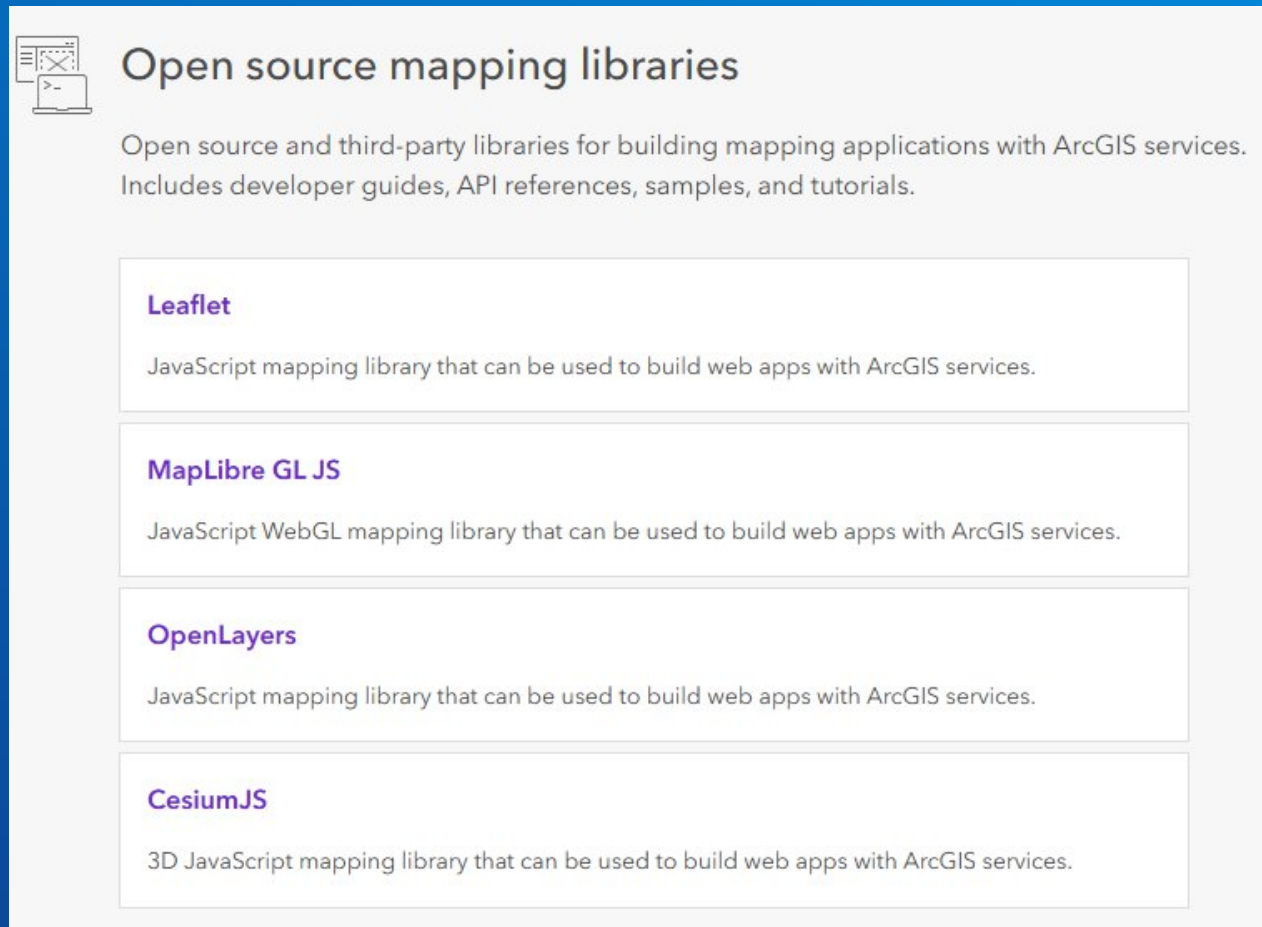
MapLibre

# Esri’s open-source ecosystem

- Javascript/Typescript npm packages

- **ArcGIS REST JS**
- **MapLibre ArcGIS plugin**
- Leaflet ArcGIS plugin (Esri Leaflet)
- CesiumJS, OpenLayers support

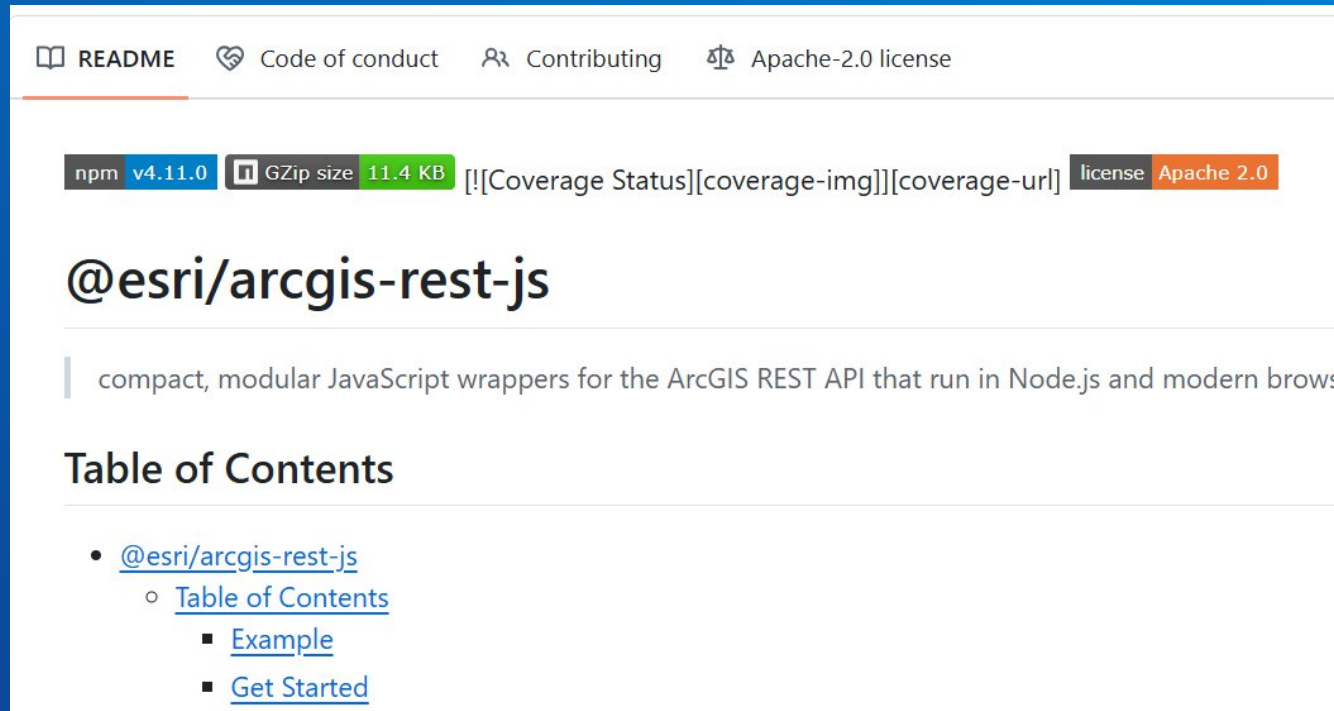
- More plugins coming soon!



# ArcGIS REST JS

@esri/arcgis-rest-js

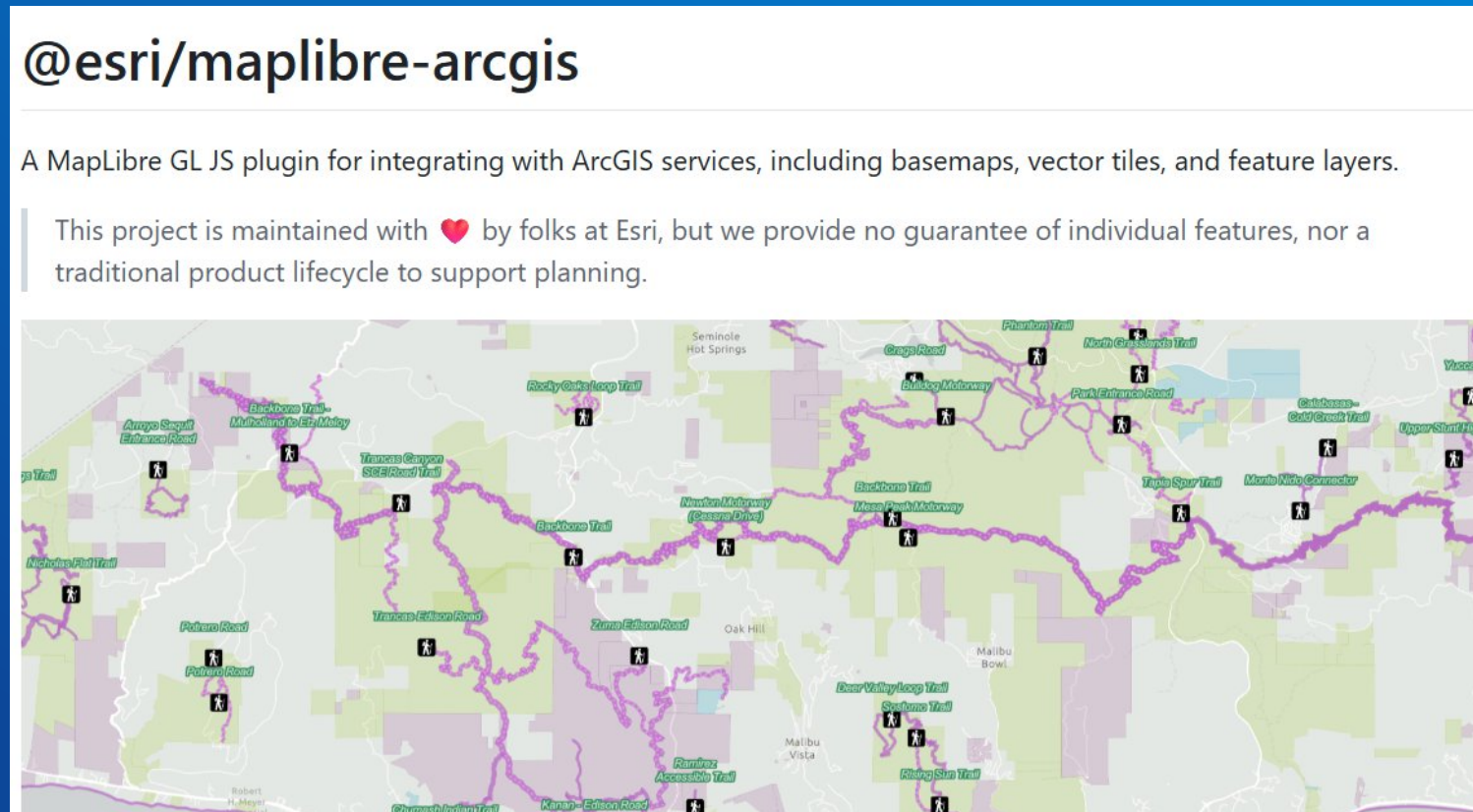
- Typescript monorepo containing 10 packages
- REST API helpers – data services, portal authentication, etc
- Core re-usable logic for other packages
- Also powers ArcGIS Hub!
- LINK



# MapLibre ArcGIS plugin

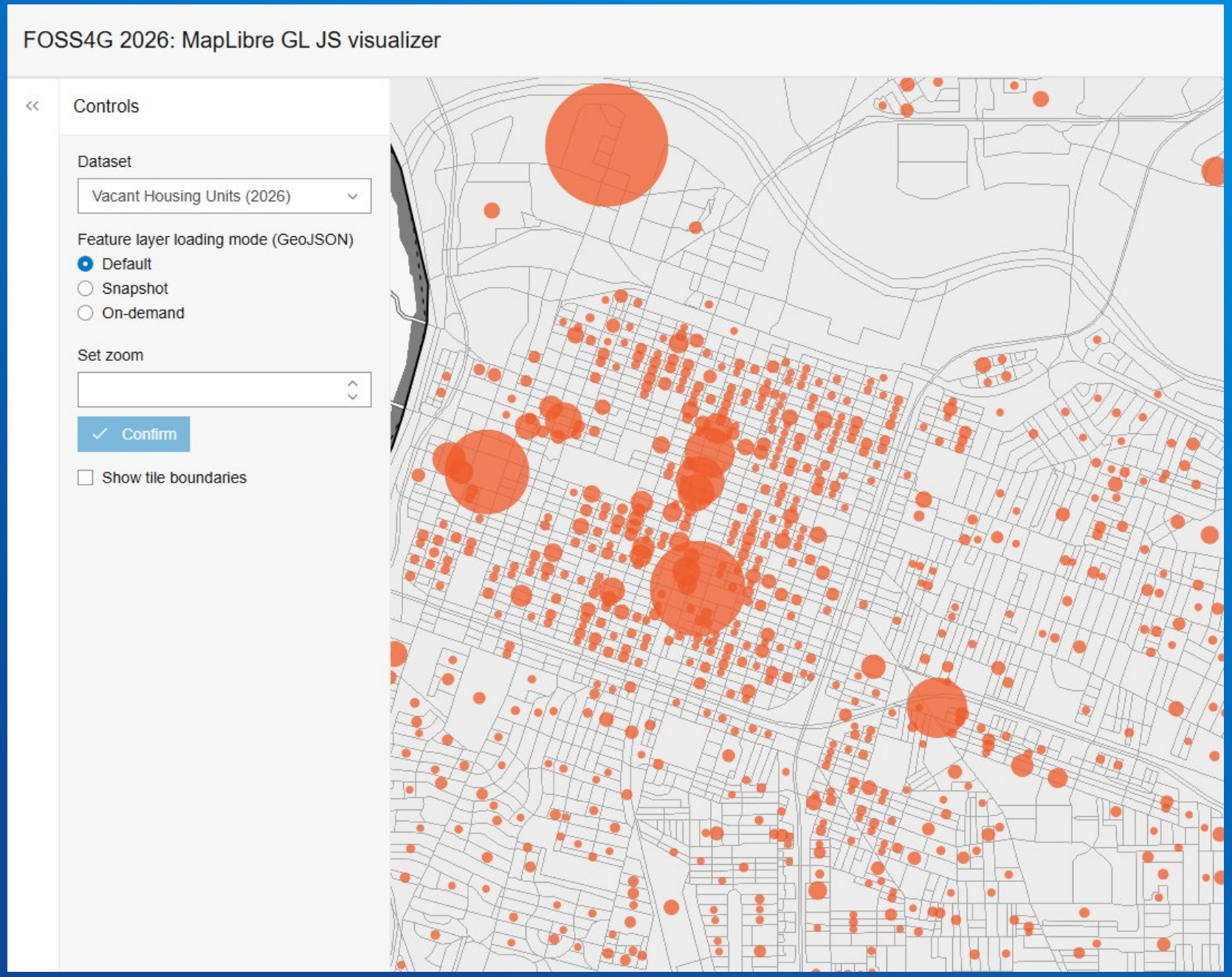
@esri/maplibre-arcgis

- Data service integration with MapLibre GL JS
- Basemap services
- Feature services
- Vector tile services
- Attribution, authentication, etc
- LINK



# Demo

- LINK

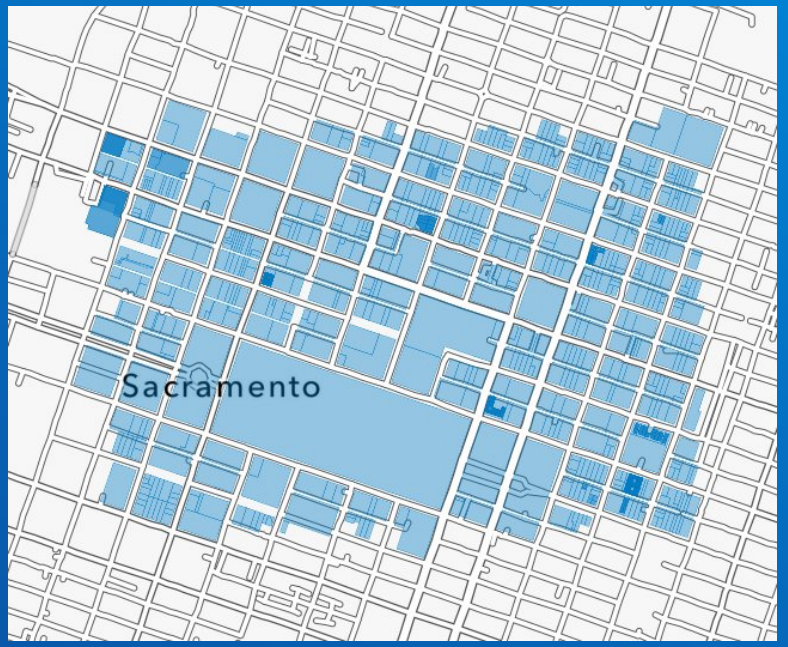

FOSS4G 2026: MapLibre GL JS visualizer
Controls
Dataset
Vacant Housing Units (2026)
Feature layer loading mode (GeoJSON)
Default
Snapshot
On-demand
Set zoom
Confirm
Show tile boundaries

# Optimizing GeoJSON queries

# Problem: GeoJSON

MapLibre ArcGIS

- Esri feature services only serve GeoJSON
  - *Spatially Optimized Parquet service also available, but distinct from feature services
- GeoJSON is unoptimized
  - Large file sizes
  - Slow loading times
- Most FeatureService optimizations require “EsriJSON” format
- Reality: **GeoJSON is still hugely popular**



# Solution: FeatureLayer class

MapLibre ArcGIS

- Load feature services in MapLibre
- Lessons learned from ArcGIS Maps SDK for JS
- Huge amount of optimizations
  - Multiple loading strategies
  - Advanced caching strategy
  - Network transfer optimizations
  - Converts EsriJSON to GeoJSON
- “The secret sauce”






# Loading strategy: Snapshot mode

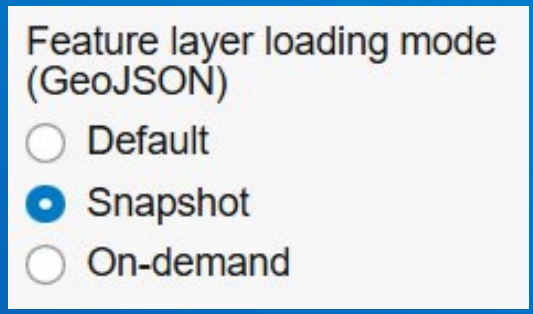
MapLibre ArcGIS

- **Load everything at once**

- Triggers on initial render
- Up to 8,000 features
  - Bypasses “2,000 feature limit”

- Easy to implement

- Real advantages for small datasets

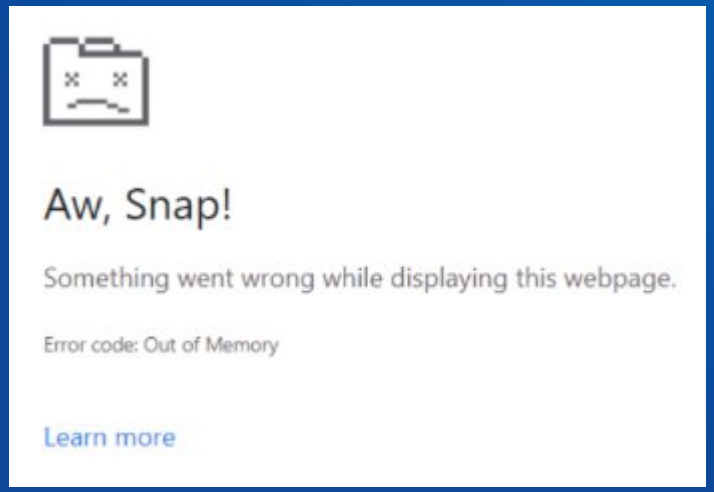


# Snapshot mode pros and cons

MapLibre ArcGIS

- + Avoids unnecessary requests
- + Real advantages for small datasets
- + Ideal situation: Get it out of the way early

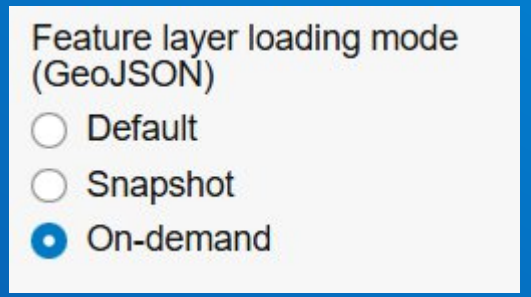
- - Blocks first contentful paint (FCP)
- - Parcels:



# On-demand loading strategy

MapLibre ArcGIS

- **Load tiles as you pan and zoom**
- Cache the current view and a bit more
- Optimize geometries at different zoom levels
  - Quantization
- Enables truly large datasets



# Quantization

MapLibre ArcGIS

- Handled by FeatureServer
- Simplify geometry of features

- Tolerance parameter: Request different versions at different zoom levels

- API reference

# More on-demand optimizations

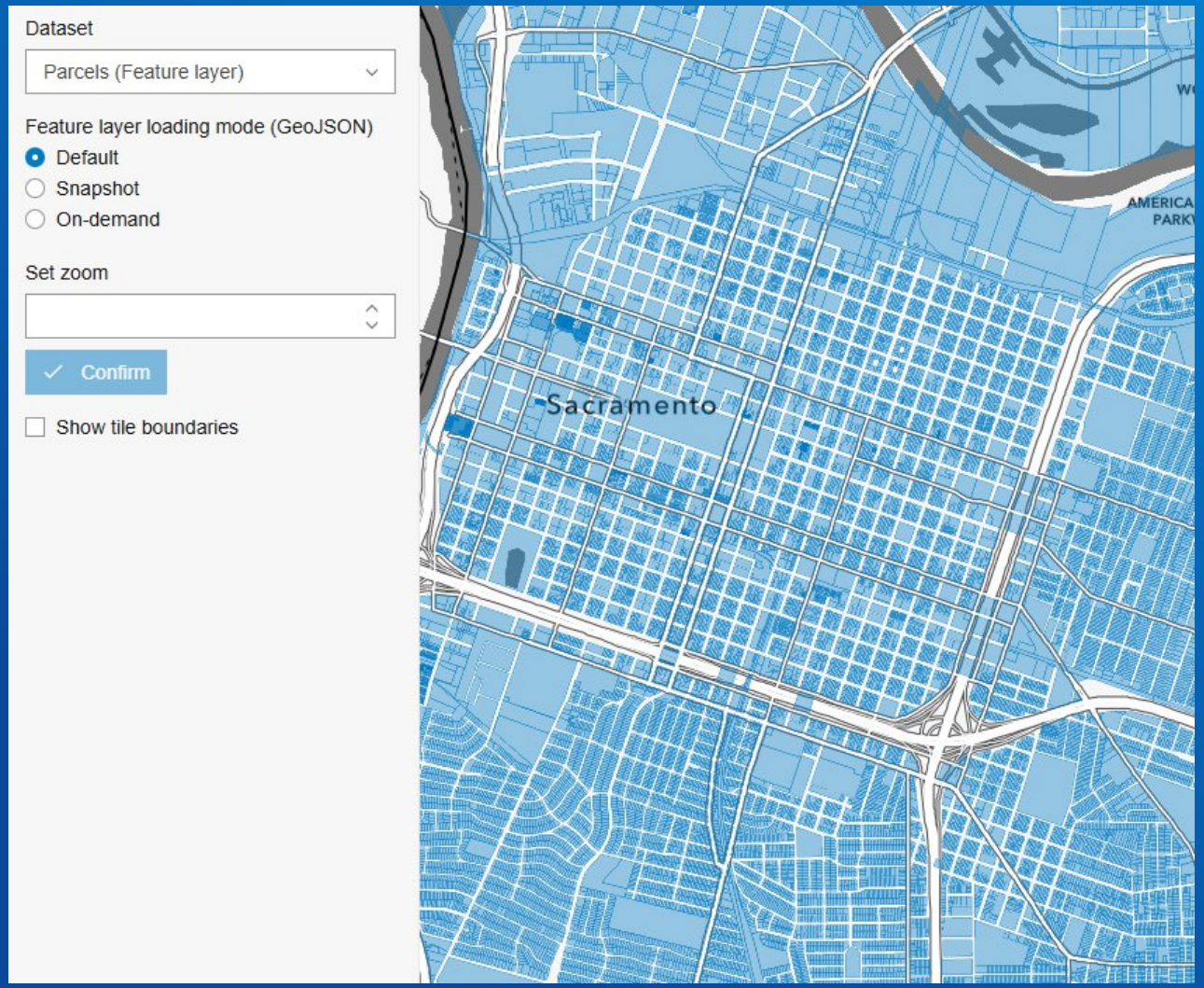
MapLibre ArcGIS

- Client caching strategies
  - Lagging window method
  - Current zoom and extent
  - Extent-based expiration
  - Lagging window method
- AbortController to deduplicate requests
- FeatureService: Server-side cache

- Hardcoded overflow prevention: 350,000 features on-screen at once
- Dependent on hardware specs

# Demo: Snapshot vs On-demand

## MapLibre ArcGIS



# Another optimization:  PBF network requests

ArcGIS REST JS

- Request all GeoJSON as highly compressed .pbf files
- Decode on client
- ~10x less bandwidth than GeoJSON

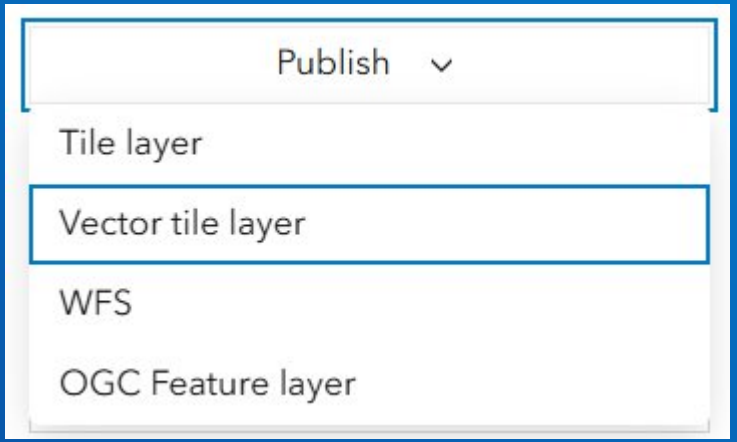
- From feature services: PBF containing EsriJSON :\
- We wrote our own parser: @esri/arcgis-rest-feature-service

- MapLibre ArcGIS workflow:
  - Query EsriJSON PBF -> convert EsriJSON to GeoJSON -> render GeoJSON

# The best optimization?

MapLibre ArcGIS

- Stop using GeoJSON...
- Publish vector tiles instead.
  - Identical attributes and geometry
- Or, use Parquet Feature Layer



# Conclusion: Feedback?

## MapLibre ArcGIS

- What do you want to see in MapLibre ArcGIS?

- Slides and app: https://github.com/gowin20/foss4g-2026-maplibregl-data-visualization




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